Canvas #01/10

Canvases:

PROBLEM SPACE

SOCIAL ROBOT CO-DESIGN CANVASES

What problem are you solving?

USER

	Group(s)	Characteristics	Needs
	Who are the users? Are there supporting users? For example: students and teachers.	What are the users like?	What do the users need?
primary users	ED patients using pre-triage	Wide variation in age, mobility, language, health literacy, digital literacy and clinical conditions who may be anxious, distressed or in pain.	Simple instructions, accessible interactions, privacy and clear information about what their next steps are.
secondary users	Triage nurses/ ED clinical staff	Clinically trained staff working in a busy, interruption-heavy environment	Reliable patient information/vitals, immediate escalation of red flags and efficient handoff in the clinical window.

Goal(s)

What do the primary and secondary users want to accomplish?

	SHORT-TERM	LONG-TERM
primary users	Complete pre-triage safely and efficiently; receive clear guidance through symptom intake and vital collection; understand what happens next	Experience shorter intake delays; improved accessibility and a smoother pre-triage experience while maintaining access to human clinical support.
secondary users	Receive organized patient information; vital signs and red flag alerts from robot to support timely nurse assessment.	Reduce repetitive intake workload; improve patient-flow efficiency and support earlier identification of high-risk patients while maintaining clinician oversight.

ROBOT

Task(s)

What task does the robot perform?

SHORT-TERM	LONG-TERM
Detects and greets patient; support language/accessibility selection; conduct rapid safety screen; guide patient to assessment position; collect symptom information; guide vital-sign measurements; transmit information to nurse; provide next step/queue guidance.	Safely navigate the designated pre-triage area; support multiple-patient intake workflows; integrate with hospital systems and return to its ready position for subsequent patients.

Advantages

What advantages does using a robot bring (compared to a computer or human)?

Social skills	User's emotional response	Personalization	Precise tasks
Greets and verbally guides patients	Uses calm, non-alarming communication.	Adapts language/accessibility guidance.	Guides structured intake consistently

Data collection with sensors	Mobility	Environment manipulation	Connection to systems
Proximity sensors + connected vital-sign devices	Moves within the designated pre-triage area and guides patients.	Minimal because it does not physically manipulate patients/ environment.	Sends collected information to nurse dashboard.

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ETHICAL CONSIDERATIONS

SOCIAL ROBOT CO-DESIGN CANVASES

Consider potential ethical problems, and potential solutions –both from the user's and robot's perspectives.
Consider the boxes to be guidelines: you don't need to fill each one.

Physical safety

Machines can pinch or crush the user. How is this mitigated?

	PROBLEM	SOLUTION
USER	Patients may collide with, trip or be unable to safely follow the robot.	Provide clear movement cues, accessible routes and immediate staff assistance when needed.
ROBOT	Movement in a crowded ED creates collision risks with patients, staff or other equipments	Use proximity/obstacle sensor, limit area/ the robot is allowed to travel, low travel speeds, automatic stopping and manual emergency stop.

Data security

Is the robot in a unique data collection position? How is the user's data protected?

	PROBLEM	SOLUTION
USER	Patients may disclose sensitive symptoms where others can see or hear them.	Provide touchscreen/private-response options and clearly explain how the information will be used.
ROBOT	Robot collects identifiable symptoms, vitals and potentially voice/interface data.	Encrypt transmissions, minimize stored data, restrict access and automatically clear patient information after handoff.

Transparency

How does the robot share an accurate perception of its abilities, intentions and constraints, so the user can evaluate their trust in it?

	PROBLEM	SOLUTION
USER	Patients may believe the robot is a clinician or is independently diagnosing/triaging them.	Clearly state that the robot performs pre-triage data collection and that a nurse makes the final clinical decision.
ROBOT	Human-like communication could make its capabilities appear greater than they actually are.	Communicate limitations clearly and identify when information is being sent to a human clinician.

Equality across users

Robots' algorithms can be biased. A robot's appearance could reinforce harmful stereotypes. What are potential issues?

	PROBLEM	SOLUTION
USER	Language, disability, age, health literacy or digital literacy may affect ability to use the robot.	Offer language selection, large readable controls, accessibility options and immediate human assistance.
ROBOT	Sensors, speech recognition or AI screening may perform differently across patient populations.	Test across diverse users, provide alternative input methods and escalate uncertain readings to staff.

Emotional consideration

People have been shown to form emotional attachments to robots, as if they were alive. Is this a potential problem?

	PROBLEM	SOLUTION
USER	Sick, anxious or distressed patients may find robotic interaction intimidating or impersonal.	Use calm language, simple instructions and always provide an option to request human asisstance.
ROBOT	Human-like behaviour could encourage inappropriate trust or emotional attachment.	Maintain a professional, supportive interaction without pretending to have emotional attachment.

Behaviour enforcement

People could transfer their inappropriate behaviour, such as rudeness, from robots to humans. How is this mitigated?

	PROBLEM	SOLUTION
USER	Patients may follow robot instructions even when uncomfortable, confused or becoming more unwell.	Allow patients to stop the interaction and request a nurse at any time.
ROBOT	Robot could continue routine questions despite detecting urgent symptoms or abnormal measurements.	Urgent bypass where it stops routine pre-triage, alert clinical staff and direct the patient to immediate human assessment.

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DESIGN GUIDELINES

SOCIAL ROBOT CO-DESIGN CANVASES

What things are important to consider in the robot's design?

Advantage guidelines

What advantages can the robotic solution have?
Think back to what you defined in the solution space canvas.

- Automate repetitive pre-triage data collection.
- Collect structured symptoms and connected vital signs.
- Reduce duplicate data collection by nurses.
- Guide patients consistently through pre-triage.
- Use mobility to support patients within the designated intake area.
- Transfer collected information directly to the nurse dashboard.

Ethical guidelines

What ethical considerations does the robot have?
Think back to what you defined in the ethics canvas.

- Clearly identify the robot as a pre-triage support system, not a clinician.
- Obtain patient cooperation/consent before collecting information.
- Protect privacy during symptom and vital-sign collection.
- Provide human assistance at any time.
- Use an urgent-bypass pathway for red-flag symptoms.
- Never allow the robot to make the final triage decision.

ROBOT DIMENSIONS

Environment guidelines

What should the robot's context be like?
For example:
- If users are especially vulnerable, should it optimize for support?
- If the robot is part of a strict process, should it optimize for efficiency and security?

- Operate only within the designated ED/pre-triage area.
- Navigate safely around patients, staff, wheelchairs and equipment.
- Maintain low movement speeds and safe stopping distances.
- Function in noisy, crowded conditions.
- Allow sufficient space around the vital-sign station.
- Use surfaces/components suitable for routine infection control.

Form guidelines

What guides the design of the robot's outward qualities?
For example:
- Should the robot be designed to appear especially approachable, or more industrial?
- Should it be simple, or detailed?

- Clearly appear to be a healthcare-assistance device.
- Use a stable mobile base with visible sensors.
- Position touchscreen at an accessible height.
- Keep vital-sign devices clearly identifiable and reachable.
- Include a visible system-status indicator.
- Avoid an overly human-like appearance that could imply clinical authority.

Interaction guidelines

What guides the design of interaction?
For example:
- Is the interaction multimodal, or is one modality optimized for efficiency?
- Should the user feel empowered and lead the interaction, or does the robot provide safety via leadership?
- Is the goal of the interaction to complete a task, or explore?

- Use touchscreen plus spoken/visual guidance.
- Begin with language and accessibility options.
- Give one simple instruction at a time.
- Allow patients to repeat, correct or go back.
- Provide a clearly visible Request Nurse/Help option.
- Confirm important information before submission.
- Support alternatives to voice input in the noisy ED.

Behaviour guidelines

What guides the design of the robot's behaviour?
For example:
- Should behaviour be simple, or sensitive to context?
- Does the robot have internal drivers, or does it react to external stimuli?
- Does the robot have social skills?

- Detect approaching patients without unexpectedly pursuing them.
- Greet patients calmly and identify its purpose.
- Move slowly and announce movement before navigating.
- Stop automatically when an obstacle/person is detected.
- Escalate red flags immediately to clinical staff.
- Stop routine questioning during urgent escalation.
- After handoff, clear the session and return to its ready/home position.

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ROBOT DESIGN MVP

SOCIAL ROBOT CO-DESIGN CANVASES

It's time to design your robot MVP (Minimum Viable Product)! Remember the guidelines you defined.

Where and when

What place?
What time of day?
Does the place or time change?

Designated Emergency Department entrance/pre-triage area. Used when eligible patients arrive and before formal nurse-led clinical triage.

Robot's role

Is the robot a friend? Teacher? Helper? Something else?

Pre-triage assistant. Greets and guides patients, collects structured symptoms and vital signs, identifies potential red flags and transfers information to clinical staff.

Draw a picture

What does the robot look like?
Is it attached to something?
Does it move around?
Can its appearance be modified?

See mockup design

Personality

Does the robot have specific characteristics?
Does it have emotional states, or needs?

Calm, respectful, reassuring and professional. Uses simple language and supportive instructions without pretending to be human or a clinician.

Context-based behaviour

What external and environmental factors affect behaviour?
What data is used to adapt to context?

Adapts language/accessibility options, responds to patient input and sensor readings, stops for obstacles, escalates red flags and provides different next-step guidance according to workflow status.

Connection to systems

Is the robot connected to external systems, such as software, databases, or other robots?
How does it use these systems?

Connected to vital-sign devices, CARISURG AI and the nurse-facing HCI dashboard. Securely transfers symptoms, measurements, timestamps and preliminary risk alerts for clinician review.

Interaction modalities

What modalities are inputs to the robot? What modalities does the robot output?

INPUT

- [x] voice
- [ ] sounds
- [ ] gestures
- [x] movement
- [x] touch
- [ ] smell
- [ ] facial expressions
- [x] screens
- [ ] lights
- [x] other: proximity sensors + vital-sign devices

OUTPUT

- [x] voice
- [x] sounds
- [ ] gestures
- [x] movement
- [ ] touch
- [ ] smell
- [ ] facial expressions
- [x] screens
- [x] lights
- [ ] other

Interaction flow

Describe the most important interaction of the robot.
Note: only fill the bottom row if your robot is teleoperated.

	BEFORE	DURING	AFTER
USER	Arrives/joins pre-triage queue and approaches CARISURG.	Selects language/accessibility needs, provides symptoms and completes guided vital-sign measurements.	Receives confirmation and queue/next-step guidance; proceeds to nurse assessment.
ROBOT	Detects patient, activates, greets them and explains its role.	Guides intake and vitals, monitors for red flags and securely transfers collected information.	Confirms handoff, directs patient appropriately, clears session and becomes available for next patient.
ROBOT OPERATOR (optional)	Monitors availability/status as required.	Receives urgent escalation if red flags or robot/sensor problems occur.	Nurse reviews collected information and completes clinical triage.

THRI paper:

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ENVIRONMENT
SOCIAL ROBOT CO-DESIGN CANVASES

What is the robot's context of operation?
You can use the "Ecosystem" canvas to dive deeper into this topic.

Where
What place?
Does it change?

Designated ED entrance/pre-triage area with access to assessment seating. Robot operates within a controlled zone containing patients, staff, wheelchairs, medical equipment and other moving obstacles.

User(s)
Who is using the robot?

Primary: Eligible ED patients completing pre-triage, with varied ages, mobility, language, health literacy, accessibility needs and clinical conditions.

When
What time of day?
Does it change?

Used during ED operating hours when eligible patients arrive for pre-triage. Demand may fluctuate from individual arrivals to periods when several patients arrive simultaneously.

Secondary user(s)
Are there secondary users?
E.g. teachers that help students use a robot.

Triage nurses and ED staff who receive collected information, respond to urgent alerts, assist patients when necessary and oversee robot operation.

Data collection
Does the robot collect data from its environment?
How is it stored?

Collects patient check-in information, chief complaint, symptoms, accessibility/language selections and vital signs. Presence/navigation sensors collect environmental information required for safe operation. Clinical data are securely transmitted to CARISURG and cleared from the robot after handoff where appropriate.

TRADE-OFF:
More data collection requires more attention to data security.

Simultaneous users
How many users should be able to use the robot simultaneously?

TRADE-OFF:
More simultaenous users requires a more sophisticated robot.

one — few — several

One active patient per interaction; multiple patients may wait in the pre-triage access queue.

External sensors and actuators
Does the robot use external sensors?
Does it have external actuators, such as lights or limbs?

Proximity/obstacle sensors and camera/presence sensing support safe navigation and patient detection. Connected BP monitor, pulse oximeter and thermometer provide vital-sign measurements. Mobile base enables controlled movement.

Connection to systems
Is the robot connected to external systems, such as software, databases, or other robots?
How does it use these systems?

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THRI paper:

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FORM

SOCIAL ROBOT CO-DESIGN CANVASES

What are the robot's outward qualities? If an existing robot is used, are its qualities modified?

Draw a picture

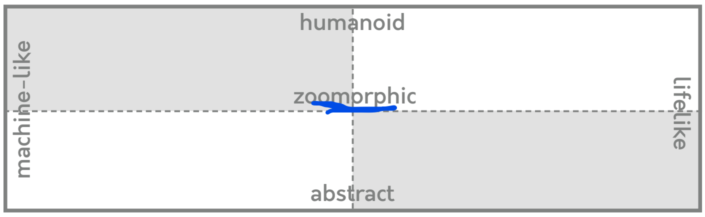
What does the robot look like?
Is it attached to something?
Does it move around?
Can its appearance be modified?

See mockup design

Appearance

Is the robot more machine or lifelike? Is it human-shaped, animal-shaped, or abstract?

TRADE-OFF:
Robots that appear more human and lifelike are expected to be more sophisticated in features.



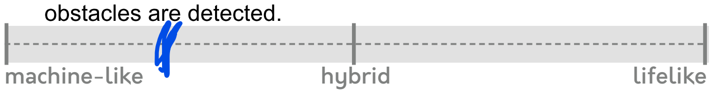
Size

How big is the robot?

Character of movement

What is the robot's movement like?

Slow, predictable and controlled movement. Robot announces movement, maintains safe distance and automatically stops when obstacles are detected.



Voice & sounds

Does the voice have a gender or an age? What are pitch, speed and prosody like? Is the voice always the same?
Does the robot make sounds: music, "beep"s, animal noises?
When are these sounds heard?

Calm, neutral and professional voice using simple instructions. Adjustable volume supports the noisy ED environment. Brief tones indicate confirmation, warnings or assistance requests; unnecessary sounds are avoided.

Mobility

Does the robot move across space? Does it move in place?

Wheeled mobile base operates within a designated pre-triage zone. Robot can approach its interaction position, guide patients to the assessment area and return to its home/ready position. Proximity and obstacle sensors support safe navigation.

Visual cues

Does the robot have expressions, lights, a screen or other visual elements?

Touchscreen provides step-by-step instructions, language/accessibility options and patient queue/next-step information. Status light communicates available, in use, assistance required or urgent escalation states. Icons accompany colour cues for accessibility.

Touch & smell sensations

Is the robot soft or rough, warm or cold?
How does the robot smell?
Touch and smell are especially important in close interactions.

THRI paper: